

# OSAMA SADEK

## Senior Full-Stack Developer & AI Automation Engineer

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Portfolio: [osamasadek.pythonanywhere.com](https://osamasadek.pythonanywhere.com) | Open to Remote Worldwide

## PROFESSIONAL SUMMARY

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Senior Full-Stack Developer and AI Automation Engineer with 11+ years of software engineering experience building scalable, production-grade web applications, cross-platform mobile systems, and AI-powered automation workflows. Expert in Python, FastAPI, React, Next.js, TypeScript, Flutter, and LLM integration (GPT-4, Gemini, Claude, RAG architectures). Proven track record of architecting 8+ production applications serving 500+ daily active users and scaling APIs to handle 10,000+ daily requests with 99.5% uptime. Reduced document processing time by 70% through intelligent AI agents and RAG pipelines. Seeking remote opportunities.

## TECHNICAL SKILLS

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- **AI & Automation:** Large Language Models (GPT-4, Gemini, Claude), LangChain, RAG Systems, Autonomous Agents, Prompt Engineering, Vector Databases (ChromaDB)
- **Frontend & Mobile:** React.js, Next.js, TypeScript, JavaScript (ES6+), Tailwind CSS, Flutter, Dart, Offline-First Architecture
- **Backend & APIs:** Python, FastAPI, Node.js, Express.js, RESTful APIs, WebSockets, Microservices, Asynchronous Processing
- **Databases & Tools:** PostgreSQL, SQLite, Prisma ORM, SQLAlchemy, Docker, Git, GitHub Actions, CI/CD, Linux

## PROFESSIONAL EXPERIENCE

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### Senior Full-Stack & AI Automation Engineer (Freelance / Remote) 2014 – Present

- Architected, developed, and deployed full-stack web and mobile systems integrated with frontier LLMs and vector search engines.
- Engineered scalable REST APIs with FastAPI and Node.js achieving 99.5% uptime while managing high concurrency.
- Implemented RAG architectures using vector embeddings, reducing manual legal and industrial document analysis times by 70%.
- Built responsive SPAs and SSR solutions using Next.js, React, and TypeScript alongside cross-platform Flutter mobile apps with offline-first sync.
- Managed end-to-end SDLC using Git, Docker, and automated CI/CD pipelines for high-reliability software deployment.

## FEATURED PROJECTS

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### Smart-IP Auditor Engine – AI Patent Analysis Platform

Stack: Python, FastAPI, GPT-4, Gemini, React.js, PostgreSQL, RAG

- Engineered an AI legal-tech platform using RAG for semantic similarity analysis, cutting review duration by 70% with 92% accuracy.

### Spotter AI – Full-Stack Logistics Route Calculator

Stack: Next.js, TypeScript, Python, FastAPI, PostgreSQL, Google Maps API

- Developed an enterprise route planning engine improving freight calculation efficiency by 60%.

### Assiut Maintenance Digital System – Enterprise Platform

Stack: Python, FastAPI, React.js, SQLite, Docker

- Digitized 42 operational workflows into a unified central dashboard serving 50+ concurrent engineering users.

### Electricity Theft Reporting App – Mobile Field Inspection Solution

Stack: Flutter, Dart, FastAPI, SQLite, Offline-First Architecture

- Built a mobile field app enabling offline video/data recording, cutting incident documentation time from 2 hours to 15 minutes.

## ADDITIONAL EXPERIENCE & DOMAIN EXPERTISE

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### Middle Egypt Electricity Distribution Company | Assiut, Egypt 2007 – 2024

#### Senior Infrastructure & Digital Transformation Manager

- Led cross-functional technical teams of 100+ personnel overseeing grid operations and digital workflow modernizations.

## EDUCATION

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### Assiut University | Egypt

Bachelor of Science in Electrical Engineering (Power & Machines) | Grade: Excellent | 2005

## CERTIFICATIONS & HIGHLIGHTS

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- Back-End AI Engineering Intern – FlyRank AI Internship (Accepted 2026)
- Gorilla Problem Solving Assessment – 97th Percentile (2025)

## LANGUAGES

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- **Arabic:** Native
- **English:** Professional Working Proficiency (Fluent)